

DP Computer Science: A 4.1.2 Machine Learning Hardware

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- **Compare and contrast** the roles of CPUs and GPUs in machine learning tasks
- **Describe** the function and optimal use cases for specialised hardware, including **TPUs, ASICs, FPGAs, and Edge Devices**
- **Analyse** a given ML scenario and recommend an appropriate hardware configuration, justifying the choice based on **scalability, processing power, cost, and the distinction between training and inference**

Key Vocabulary

Term	Definition
CPU (Central Processing Unit)	General-purpose processor; excels at sequential, complex tasks; limited parallel processing
GPU (Graphics Processing Unit)	Processor with thousands of cores; excels at parallel processing; ideal for ML training
TPU (Tensor Processing Unit)	Google's custom ASIC designed specifically for neural network computations
ASIC (Application-Specific Integrated Circuit)	A chip designed for a specific purpose; highly efficient for that task
FPGA (Field-Programmable Gate Array)	A chip that can be reprogrammed after manufacturing; flexible hardware
Edge Device	A device that performs ML inference locally (e.g., smartphone, IoT device)
Training	The process of teaching a model using large datasets; computationally intensive
Inference	The process of using a trained model to make predictions; faster, less intensive
Cloud Computing	Using remote servers for computation; scalable, pay-as-you-go
HPC (High-Performance Computing)	Supercomputers for intensive computation; used for large-scale ML research
Parallel processing	Performing many calculations simultaneously
Throughput	The amount of work completed per unit of time

Approaches to Learning (ATL) Elements

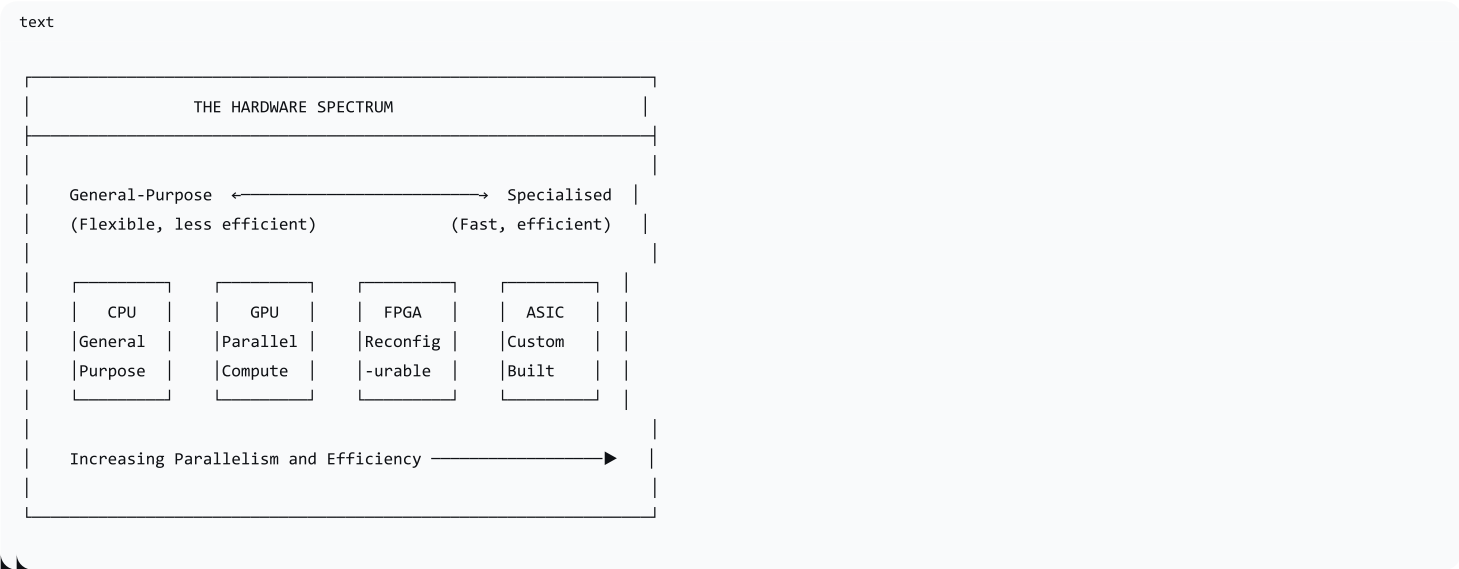
Skill	Description
Thinking (Analysis & Evaluation)	Analysing trade-offs between hardware solutions; evaluating appropriate configurations
Communication	Articulating and defending hardware recommendations; using analogies to explain technical concepts
Research	Drawing on research from previous lesson to inform hardware choices

2. From ML Models to Hardware: The Connection

Recap: From Lesson 4.1.1

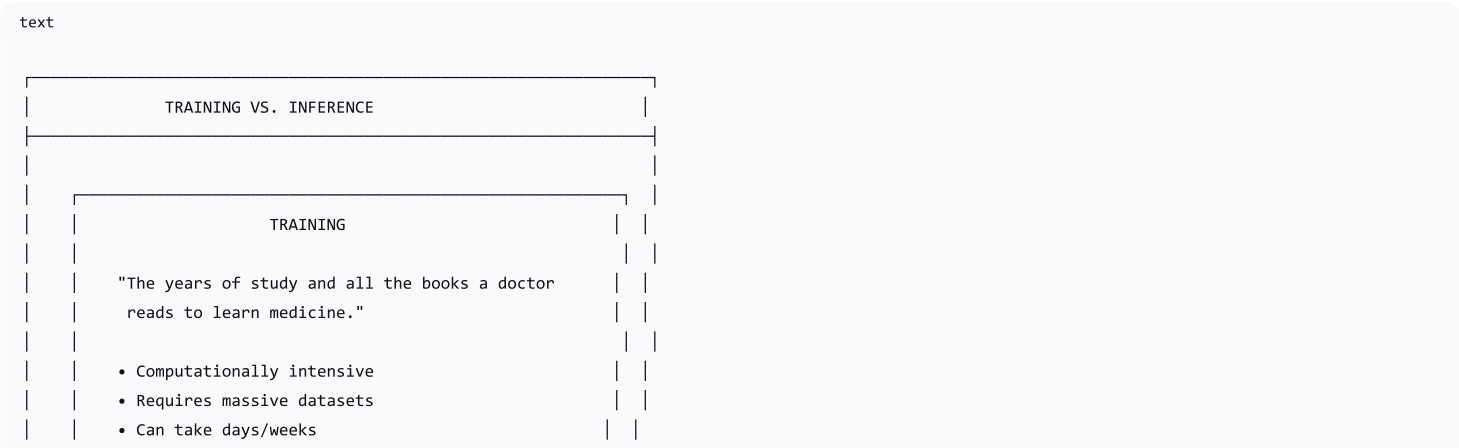
“In the previous lesson, we explored different ML approaches. But once we decide on an approach, we need to run it on something—hardware!”

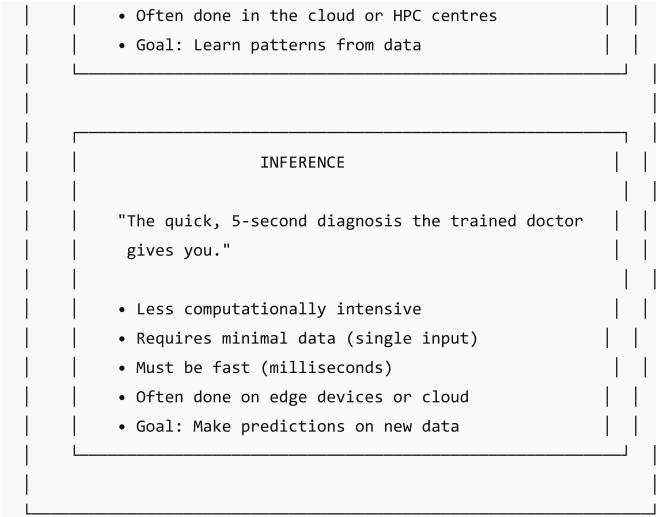
The Hardware Spectrum



3. Training vs. Inference: The Critical Distinction

The Two Phases of ML





Implications for Hardware

Aspect	Training	Inference
Compute Requirement	Very High	Low to Medium
Data Requirement	Massive	Minimal
Time Requirement	Days/Weeks	Milliseconds
Preferred Hardware	GPU clusters, TPUs, HPC	Edge devices, CPUs, mobile GPUs
Location	Cloud, Data Centres	On-device, Edge, Cloud
Cost	High (compute + storage)	Low (per prediction)

4. Comparing Hardware Types

4.1 CPU (Central Processing Unit)

Aspect	Description
Role in ML	General-purpose; manages system, coordinates other hardware, handles sequential tasks
Strengths	Flexible, good for small datasets, handles complex logic
Weaknesses	Limited parallel processing; slow for large-scale ML
Best For	Small ML projects, data preprocessing, inference on simple models

Analogy: The Master Chef

"A CPU is like a master chef who can cook any dish but can only work on one or two things at a time."

4.2 GPU (Graphics Processing Unit)

Aspect	Description
Role in ML	Parallel processing powerhouse; the workhorse of ML training
Strengths	Thousands of cores, excellent parallel processing, high throughput
Weaknesses	Expensive, power-hungry, less flexible than CPU
Best For	Deep learning training, large-scale ML, computer vision

Analogy: An Army of Cooks

| "A GPU is like an army of cooks who can each focus on a small task, working together to prepare an entire meal very quickly."

GPU vs. CPU Comparison:

Feature	CPU	GPU
Cores	Few (4-64) powerful cores	Thousands of simpler cores
Parallelism	Limited	Massive
Best For	Sequential tasks	Parallel tasks
ML Role	Coordination, simple tasks	Training, intensive computation
Example	Running a program	Training a neural network

4.3 TPU (Tensor Processing Unit)

Aspect	Description
Definition	Google's custom ASIC designed specifically for neural network computations
Role in ML	Optimised for matrix multiplication (the core of neural networks)
Strengths	Extremely fast for ML, energy-efficient, purpose-built
Weaknesses	Limited to specific tasks, only available through Google Cloud
Best For	Large-scale neural network training, Google services

Analogy: The Specialised Baking Robot

| "A TPU is like a robot built specifically for baking cakes—it's perfect for that one task and much faster and more efficient than a general-purpose chef."

4.4 ASIC (Application-Specific Integrated Circuit)

Aspect	Description
Definition	A chip designed for a specific application or task
Role in ML	Custom hardware for specific ML workloads
Strengths	Extremely efficient, high performance for specific tasks, low power
Weaknesses	Not flexible; cannot be reprogrammed

Aspect	Description
Best For	Specific ML applications, embedded systems

Analogy: A Custom Tool

"An ASIC is like a custom-built tool for a specific job—perfect for that job but not useful for anything else."

4.5 FPGA (Field-Programmable Gate Array)

Aspect	Description
Definition	A chip that can be reprogrammed after manufacturing
Role in ML	Flexible hardware that can be optimised for specific ML algorithms
Strengths	Reconfigurable, good balance of performance and flexibility
Weaknesses	Slower than ASICs, more expensive than GPUs, complex programming
Best For	Prototyping, applications requiring frequent changes, low-latency inference

Analogy: A Multi-Tool

"An FPGA is like a Swiss Army knife—you can configure it for different tasks, but it's not as perfect as a dedicated tool."

4.6 Edge Devices

Aspect	Description
Definition	Devices that perform ML inference locally (smartphones, IoT devices, cameras)
Role in ML	Run trained models for real-time inference
Strengths	Low latency, privacy, no internet required, battery-efficient (with specialised chips)
Weaknesses	Limited compute power, constrained by battery and size
Best For	Real-time applications, privacy-sensitive tasks, offline inference

Analogy: The On-Site Doctor

"An edge device is like a doctor at a small clinic who can quickly diagnose patients on the spot without sending them to the main hospital."

Examples of Edge Devices:

- Smartphones (face unlock, voice assistants)
- Smart cameras (object detection, facial recognition)
- IoT sensors (anomaly detection)
- Autonomous vehicles (real-time decision-making)

5. Hardware Comparison Summary

Hardware	Flexibility	Speed	Efficiency	Best For
CPU	Very High	Low	Low	Small models, coordination, simple tasks
GPU	High	High	Medium	ML training, deep learning
TPU	Low	Very High	Very High	Neural network training
ASIC	Very Low	Very High	Highest	Specific ML applications
FPGA	Medium	High	High	Prototyping, reconfigurable tasks
Edge Device	Medium	Low-Medium	High	Real-time inference, privacy

6. Location of Compute: Cloud vs. Edge

Where Does ML Happen?



Hardware Architectures by Scenario

Scenario	Training Location	Inference Location	Hardware
Large Language Model (GPT)	Cloud/HPC	Cloud (API)	TPUs, GPU clusters
Self-Driving Car	Cloud	On-vehicle (Edge)	GPU clusters (train), Edge GPUs (infer)
Smartphone Face Unlock	Cloud (pre-trained)	On-device (Edge)	GPU clusters (train), NPU (infer)
Medical Imaging	Cloud/HPC	Hospital (on-prem)	GPU clusters (train), Specialised servers (infer)
IoT Sensor Anomaly Detection	Cloud	On-sensor (Edge)	CPU/GPU (train), Low-power chip (infer)

7. Hardware Architect Activity

Instructions

1. You are a **Hardware Architect**
 2. Your group is assigned an ML scenario
 3. Recommend a full hardware configuration
 4. Justify your choice based on:
 - **Performance** (training vs. inference)
 - **Cost**
 - **Scalability**
 - **Power consumption**
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Scenario A: Language Model for Customer Service Chatbot

Aspect	Considerations
Task	A large language model for a customer service chatbot
Training	Requires massive compute; large dataset
Inference	Needs fast responses for many users
Budget	Large budget available

Recommended Configuration:

- **Training:** GPU cluster (multiple NVIDIA A100 or equivalent) or TPUs
 - **Inference:** Cloud-based GPUs with auto-scaling
 - **Cost Justification:** Training is a one-time cost; inference scales with users
 - **Performance Justification:** GPUs/TPUs provide the parallel processing needed for language models
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Scenario B: Self-Driving Car

Aspect	Considerations
Task	Real-time perception and decision-making
Training	Requires massive image/video datasets
Inference	Must be fast (milliseconds); on-vehicle
Power Constraints	Limited battery; must be efficient
Budget	Moderate to high

Recommended Configuration:

- **Training:** GPU cluster (cloud-based) with large storage
 - **Inference:** On-vehicle Edge GPU (NVIDIA Drive or similar)
 - **Cost Justification:** Training in cloud (cost-effective for one-time cost); inference hardware is expensive but necessary for safety
 - **Performance Justification:** Edge GPU provides real-time inference; cloud provides training capacity
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Scenario C: Medical Imaging Diagnostics

Aspect	Considerations
Task	Detect diseases from X-rays and MRIs
Training	Requires large, confidential datasets
Inference	Needs high accuracy; can be slightly slower
Data Privacy	Data must remain on-premises
Budget	High (healthcare sector)

Recommended Configuration:

- **Training:** On-premises GPU server (ensures data privacy)
- **Inference:** On-premises GPU or CPU server
- **Cost Justification:** Healthcare has strict privacy requirements; on-premises is necessary
- **Performance Justification:** GPUs provide the compute for image processing and deep learning

8. Hardware Architect Scenario Comparison

Scenario	Training Hardware	Inference Hardware	Key Justification
Chatbot LLM	GPU cluster / TPUs	Cloud GPUs (auto-scaling)	Scale for many users
Self-Driving Car	GPU cluster (cloud)	Edge GPU (on-vehicle)	Real-time, low latency
Medical Imaging	On-prem GPU server	On-prem GPU/CPU	Data privacy
IoT Sensor	Cloud CPU/GPU	Low-power edge chip	Battery efficiency
Video Streaming	GPU cluster (cloud)	Edge device (user's device)	Decentralised inference

9. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Recap & Presentations	10 min	Recap ML approaches; 1-2 groups present from Lesson 4.1.1
Lecture & Discussion	25 min	Hardware spectrum; CPU vs. GPU (analogies); specialised hardware; training vs. inference
Hardware Architect Activity	20 min	Groups analyse scenarios and recommend hardware configurations
Discussion & Consolidation	5 min	Share recommendations; discuss trade-offs

Discussion Questions

Question	Purpose
"Why are GPUs better than CPUs for ML training?"	Understand parallel processing
"What is the key difference between training and inference?"	Understand the distinction
"When would you use a TPU instead of a GPU?"	Compare specialised hardware
"Why would an application use edge devices instead of the cloud?"	Understand edge computing benefits
"How can larger models be processed using GPUs and cloud processing?"	Link to A1 (Cloud)

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Hardware Architect Activity	Evaluate ability to analyse and recommend hardware configurations
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the main difference between a CPU and a GPU?
2. What is the difference between training and inference?
3. When would you use an edge device instead of the cloud?
4. Name one specialised hardware type for ML and describe its purpose.

11. Differentiation

Level	Strategy
Support	Provide analogies and visual aids; use the "Kitchen" and "Toolbox" analogies repeatedly
Challenge	Research hardware specifications for real-world ML deployments (e.g., GPT-4 hardware)

12. Resources

- **Presentation** (Slide Deck)
- **eBook** (A4.1.2 with examples)
- **Worksheet:** Hardware scenario analysis
- **Worksheet Answers**
- **Quizlet** (A4.1.2 vocabulary flashcards)

13. Summary: Key Takeaways

Concept	Key Point
CPU	General-purpose; sequential processing
GPU	Parallel processing; ML training workhorse
TPU	Google custom ASIC for neural networks
ASIC	Custom chip for specific tasks
FPGA	Reprogrammable; flexible hardware
Edge Device	On-device inference; low latency, privacy
Training	Compute-intensive; cloud/HPC
Inference	Fast, lightweight; edge or cloud

text

KEY REMINDERS

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CPU = General purpose, sequential

✓

GPU = Parallel processing, ML training

✓

TPU = Purpose-built for neural networks

✓

ASIC = Custom hardware for specific tasks

✓

FPGA = Reprogrammable hardware

✓

Edge = On-device inference, low latency

✓

Training = Heavy compute, cloud

✓

Inference = Fast, edge

"The choice of hardware for machine learning depends on the task—training requires massive parallel processing (GPUs, TPUs), while inference often benefits from low-latency edge devices."